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1. Introduction

In this lab, we examined a two stage amplifier with CE (Common Emitter) amplifier used for gain and CC (Common Collector) amplifier as a buffer.. Only a single circuit was constructed in order to accomplish this study. This lab took place on March 26, 2025. The signed pages are verified by the TA at the end of the report in the Appendix.

2. Objectives

The lab's objective is to show the performance of a CE-CC amplifier. It requires the construction of two transistor amplifiers and the measurement of DC values (voltage and current) along with the analysis of AC parameters (voltage gain). This lab built a better understanding of the advantages of using a CC buffer in amplifier design for driving low impedance loads while maintaining signal strength.

3. Circuit Under Test

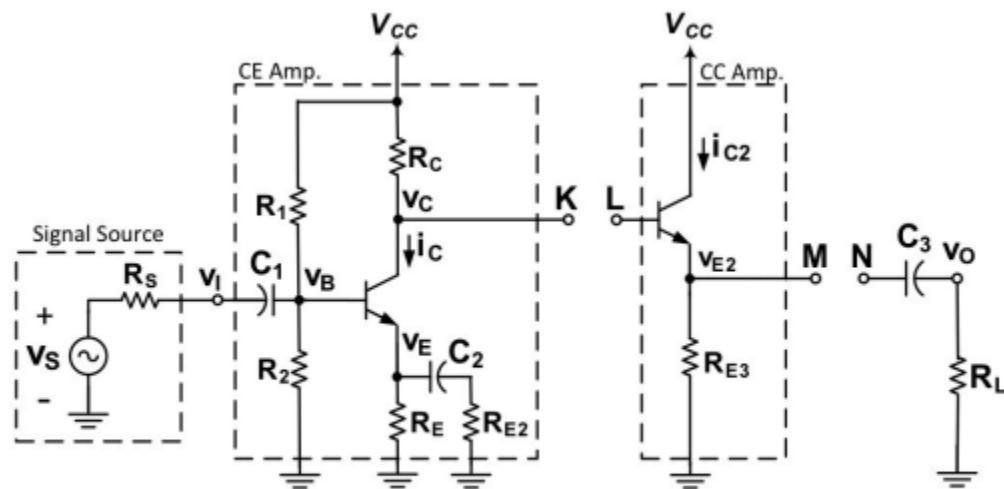


Figure 3. A circuit that can be configured to represent the CE amplifier of Figure 1 or the two-stage amplifier of Figure 2.

This circuit is a two-stage common-emitter (CE) amplifier designed to amplify AC signals. The first stage consists of a CE amplifier with a voltage divider biasing network formed by R_1 and R_2 , which sets the transistor's base voltage. The emitter resistor R_E provides stability, while the collector resistor R_C develops the amplified signal. Bumper jumpers can be located across the letters KL and MN, which are used to configure the circuit for different operation modes such as a single-stage and two-stage common-emitter (CE) amplifier.

4. Experimental Results

Amplifier	Jumpers	$V_c[V]$	$V_E[V]$	$I_c[mA]$	$V_{E2}[V]$	$I_{c2}[mA]$
CE	None	11.94	7.68	4.44	...	
Two stage	K-L	11.8	7.68	4.43	11.095	9.25

Table E1. Quiescent parameters of CE amp. of Figure 1 and two -stage amp. Of Figure 2

Amplifier	Jumpers	$V_i[V_{rms}]$	$V_c[V_{rms}]$	$V_{E2}[V]$	$V_o[V_{rms}]$	$A_{vo}[V]$
CE	None	63.4 mV	0.71	...	...	11.19

Table E2(a). No -Load AC voltages and gain of the CE amplifier of Figure 1

Amplifier	Jumpers	$V_i[V_{rms}]$	$V_c[V_{rms}]$	$V_{E2}[V]$	$V_o[V_{rms}]$	$A_v[V]$
CE	K-N	63.4 mV	0.532V	...	28.22	0.45

Table E2(b) . Loaded AC voltages and gain of the CE amplifier of Figure 1

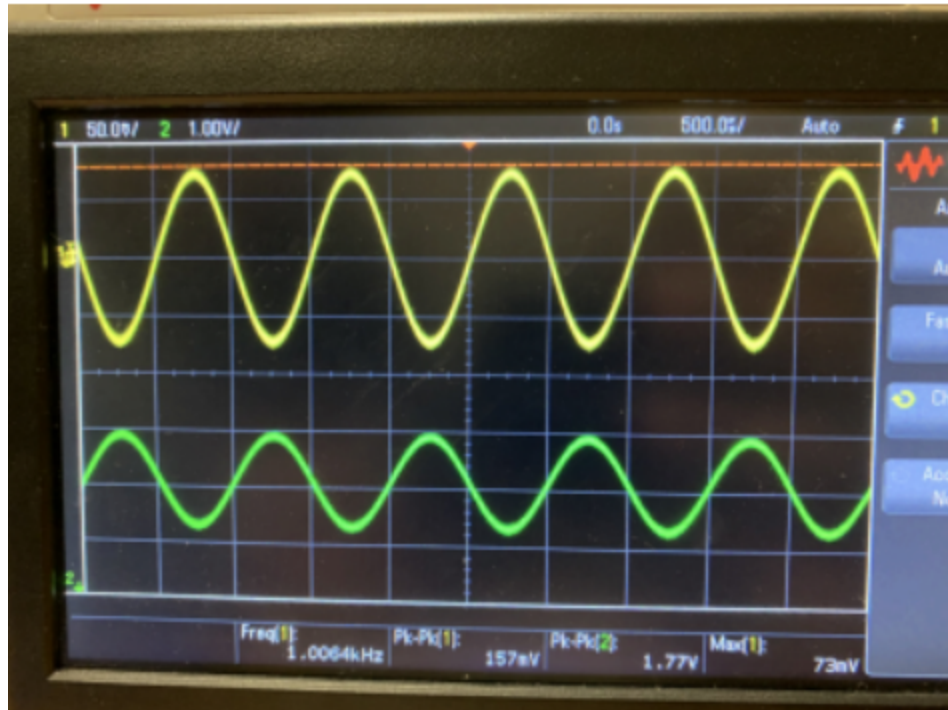
Amplifier	Jumpers	$V_i[V_{rms}]$	$V_c[V_{rms}]$	$V_{E2}[V]$	$V_o[V_{rms}]$	$A_{vo}[V]$
Two stage	K-L	63.4 mV	0.813V	0.708	...	11.17

Table E3(a). No -Load AC voltages and gain of the two- stage amplifier of Figure 2

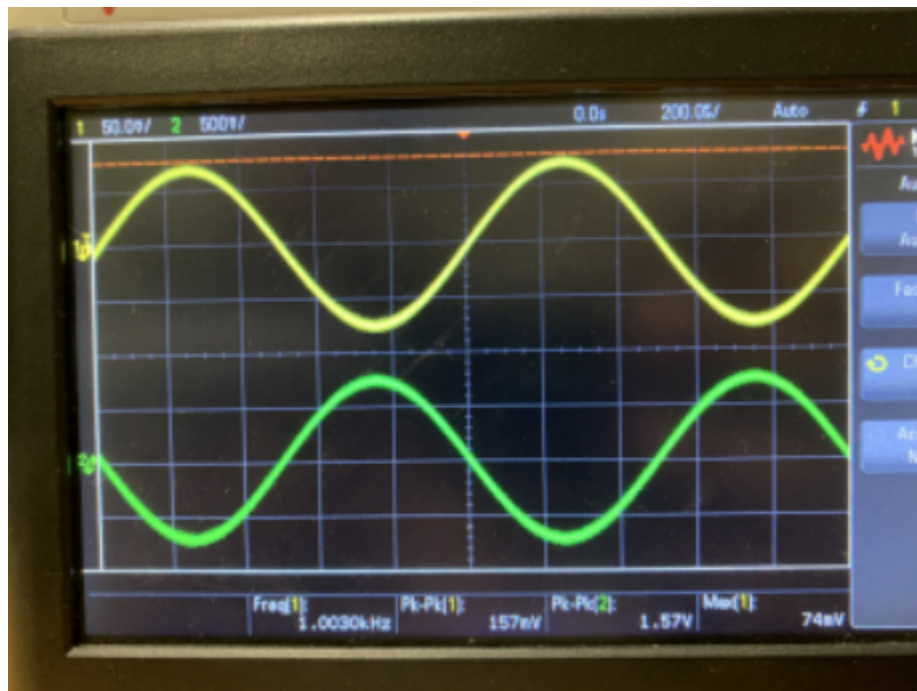
Amplifier	Jumpers	$V_i[V_{rms}]$	$V_c[V_{rms}]$	$V_{E2}[V]$	$V_o[V_{rms}]$	$A_v[V]$
Two Stage	K-I, M-N	63.4 mV	54.84mV	54.87mV	0.62V	9.78

Table E3(b). Loaded AC voltages and gain of the two-stage amplifier of Figure 2

Graph E2:



Graph E3:



5. Conclusions and Remarks

C1. Comparing the theoretical quintessential voltages of the CE amplifier with measured values.

	$V_c[V]$	$V_E[V]$	$V_{E2}[V]$
Calculated Values (from Table P1)	12	7.54	...
Measured Values (first column, Table E1)	11.94	7.68	...
Percent Error, e%	0.083	0.498	...

Table C1(a): Calculated and measured (DC) voltages of the CE amplifier of Figure 1.

	$V_c[V]$	$V_E[V]$	$V_{E2}[V]$
Calculated Values (from Table P2)	12	7.54	11.3
Measured Values (first column, Table E1)	11.8	7.68	11.095
Percent Error, e%	1.12	1.03	1.79

Table C1(b): Calculated and measured (DC) voltages of the two-stage amplifier of Figure 2.

C2. Comparing the measured voltage gains of the CE amplifier with the calculated values.

	$A_v[V]$	$A_{vo}[V]$

Calculated Values (from Table P1)	0.996	11.14
Measured Values (from Table E2(a) and Table E2(b))	0.45	11.19
Percent Error, e%	400	0.44

Table C2(a): Calculated and measured voltage gains of the CE amplifier of Figure 1.

	$A_v[V]$	$A_{vo}[V]$
Calculated Values (from Table P2)	10.97	11.14
Measured Values (from Table E3(a) and Table E3(b))	9.78	11.17
Percent Error, e%	10.8	0.27

Table C2(b): Calculated and measured voltage gains of the two-stage amplifier of Figure 2

Based on the above results the percent error for A shows that the results observed and V_0 calculated are accurate however A had an error of 7.02% which could be a result of component V variations, equipment errors, and biasing inaccuracies.

C3. The impact of the CC stage in maintaining a good overall voltage gain under heavy load.

R_{i2} = Input Impedance of CC Stage
 R_{i2} w/o load
 $R_{i2} = \beta \cdot R_E = (150)(1.2k) = 180k\Omega$
 R_{i2} with load
 $R_{i2} = \beta \cdot (R_E // R_L)$
 $= 150 \left(\frac{1.2k \cdot 0.18k}{1.2k + 0.18k} \right)$
 $= 23.5k\Omega$

The CC stage improves performance by presenting high input impedance ($R_{i2} = 23.5k\Omega$ with $R_L = 180\Omega$) to the CE amplifier which prevents excessive loading. If the CC buffer wasn't there, the CE amplifier would directly drive the load which would lower gain. The CC stage's low output impedance drives R_L which leads to a maintained near unity voltage gain at output while still maintaining CE stage's high voltage gain.

6. Appendix TA Copy

Pre Lab

E1

Amplifier	Jumpers	V_C [V]	V_E	I_C	V_{E2}	I_{C2}
CE	None	11.94	7.68		—	—
two-stage	K-L	11.8	7.68		11.095	

E2(a)

Amplifier	Jumpers	V_i	V_C	V_{E2}	V_o	A_{Vo}
CE	None	63.4mV	0.71V	—	—	$\frac{V_o}{V_i} = 11.19$

E2(b)

Amplifier	Jumpers	V_i	V_C	V_{E2}	V_o	A_V
CE	K-N	63.3mV	0.71V 0.66V 44.85mV 0.532V	—	0.642V 44.7mV 28.22mV	$\frac{V_o}{V_i} = 0.45$

E3(a)

Amplifier	Jumpers	V_i	V_C	V_{E2}	V_o	A_{Vo}
two stage	K-L	63.39mV	0.813V	0.708V	—	$\frac{V_o}{V_i} = 11.17$

E3(b)

Amplifier	Jumpers	V_i	V_C	V_{E2}	V_o	A_V
two stage	K-L-M-N	63.39mV	0.707V 54.84mV	0.707V 54.87mV	0.62V 0.62V	$\frac{V_o}{V_i} = 9.78$